



Genesis Biotech Inc.

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IGF-1 (Insulin-Like Growth Factor I)

Catalog No. PG-40001

Scale: 100 µg

Protein species: Human

Bioactivity (ED50): 1.0-3.0 ng/ml

Applications tested: cell culture

Expression Host: *E. coli*

Type: purified recombinant protein

Form: lyophilized protein

Target description

Insulin-like growth factor-1 (IGF-1) is a hormone that is secreted from the liver and other tissues in response to growth hormone. A majority of the growth promoting effects of growth hormone is actually due to IGF-1 acting on its target cells. IGF-1 stimulates proliferation of chondrocytes (cartilage cells), resulting in bone growth. It appears to be the key player in muscle growth. It stimulates both the differentiation and proliferation of myoblasts. It also stimulates amino acid uptake and protein synthesis in muscle and other tissues. In experimental animals, cardiomyocyte apoptosis inhibited by IGF-I has been demonstrated in ischemic cardiomyopathy, but it is not known whether IGF-I suppresses the apoptosis of cardiomyocytes in dilated cardiomyopathy.

Antibiotic residue

No activity of ampicillin or other antibiotic residue be detected.

Bacterial endotoxin content

Bacterial endotoxin content is not higher than 0.1ng/ug (1EU/ug)

Biological Activity

•The ED50 is defined as the effective concentration of growth factor (dose) that elicits a 50% increase in cell growth in a cell based bioassay. ED50: 1.0-3.0 ng/ml

Potential applications

1. Anti-Aging: providing the raw material needed to repair damage to the DNA and initiates cell division.
2. Increasing muscles strength.
3. Prevention of colds and flu through the promotion of wellness.
4. Prevent anemia.
5. Normalize blood pressure.
6. Reduce cholesterol.
7. Wound healing.
8. Studies demonstrate that IGF-1 augments the repair of injured nerve and muscle.

Storage

For short-term use store at 4 °C. For long-term storage keep at -20°C.

References

1. M. Sohmiya and Y. Kato. Human growth hormone and insulin-like growth factor-I inhibit erythropoietin secretion from the kidneys of adult rats. *J Endocrinol* 184(1):199-207,2005.
2. A. L. Theiss, S. Fruchtmann and P. K. Lund. Growth factors in inflammatory bowel disease: the actions and interactions of growth hormone and insulin-like growth factor-I. *Inflamm Bowel Dis* 10(6):871-80, 2004.
3. W. L. Lee *et al* Insulin-like growth factor I cardiovascular function and suppresses apoptosis of cardiomyocytes in dilated cardiomyopathy. *Endocrinology*. 140(10):4831-4840, 1999.

Comments

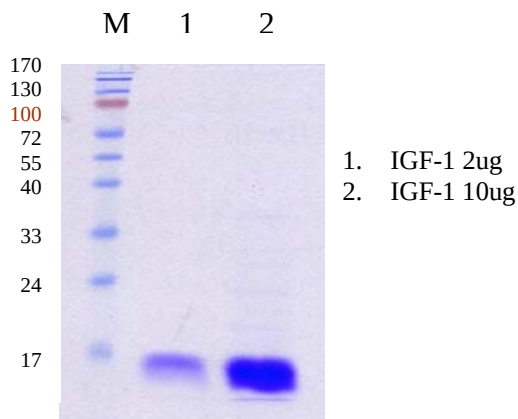
IGF-1(Recombinant human Insulin-like Growth Factor-1): Molecular Mass: 7.8kDa, composed of 71 aa. Human IGF is recombinant DNA produced and >95% pure via SDS-page.

Solubility

It is recommended to reconstitute the lyophilized IGF-1 in sterile H₂O not less than 100µg/ml, which can then be further diluted to other aqueous solutions.

Related Products

1. PDGF-B (platelet-derived growth factor B-chain) (PG-40002)
2. bFGF (basic fibroblast growth factor) (PG-40003)
3. Fibronectin (PG-40004)
4. TNF (Tumor necrosis factor) (PG-40005)



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